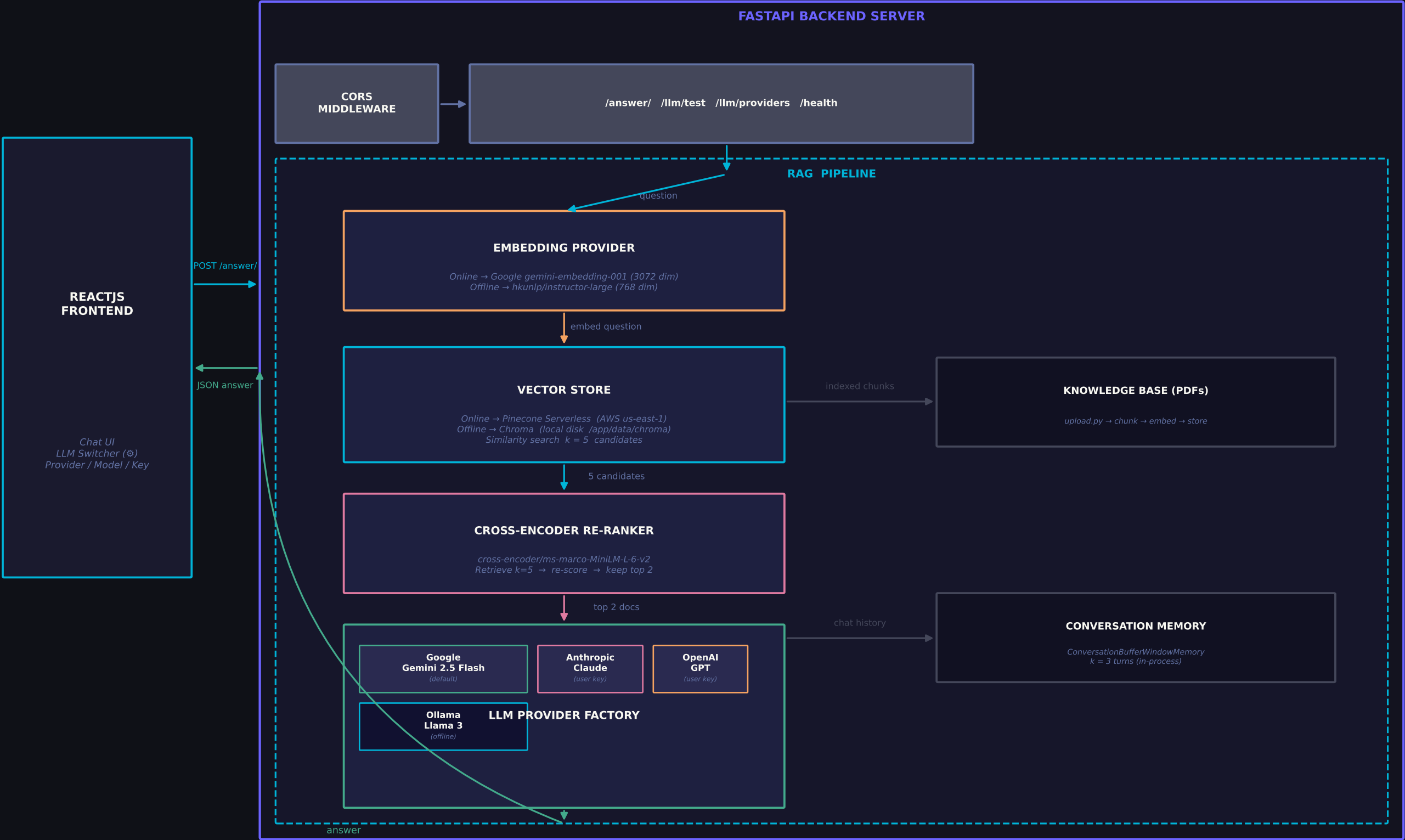


STOCKBOT — RAG PIPELINE ARCHITECTURE

Current Version | Online: Pinecone + Gemini | Offline: Chroma + Ollama



STOCKBOT — ONLINE vs OFFLINE VERSIONS

Switched entirely via environment variables — same codebase, different Docker build

ONLINE VERSION

deploy target | Dockerfile.online

Embeddings

Google gemini-embedding-001
3072 dim | cloud API

Vector Store

Pinecone Serverless
AWS us-east-1 | 3072 dim index

Default LLM

Gemini 2.5 Flash
DEFAULT_LLM_PROVIDER=google

User LLMs

Anthropic Claude / OpenAI GPT
via LLM Switcher UI

RAM Required

~300 MB
(no local model)

Internet

Required

EMBEDDING_PROVIDER=google | VECTOR_STORE=pinecone
DEFAULT_LLM_PROVIDER=google | PINECONE_API_KEY=... | GOOGLE_API_KEY=...

OFFLINE VERSION

local / air-gapped | Dockerfile.offline

Embeddings

hkunlp/instructor-large
768 dim | local HuggingFace

Vector Store

Chroma (in-process)
local disk /app/data/chroma

Default LLM

Ollama + Llama 3
DEFAULT_LLM_PROVIDER=ollama

User LLMs

Any model via Ollama
llama3 / mistral / phi3

RAM Required

~3 GB
(embedding model + Ollama)

Internet

Not required after setup

EMBEDDING_PROVIDER=local | VECTOR_STORE=chroma
DEFAULT_LLM_PROVIDER=ollama | OLLAMA_HOST=http://ollama:11434 | OLLAMA_MODEL=llama3

Re-Ranker (cross-encoder/ms-marco-MiniLM-L-6-v2) is shared across both versions